AI-Based Automated Financial Report Generation System

Companies Analysed	275 (5 sectors)
Years Covered	2019 - 2023
ML Models	XGBoost (R2=0.9650) vs LightGBM (R2=0.9254) | Best: XGBoost
AI Narrative	Gemini AI (Google)
Report Generated	2025

Key Performance Indicators

Metric	Value	Metric	Value
Avg Revenue	$2,521.8 M	Avg Net Profit	$375.9 M
Profit Margin	16.22%	Avg Debt Ratio	0.459
Avg ROA	8.16%	High Leverage	27% of companies
Best Model (XGBoost)	R2=0.9650	Model MAE	$33.7 M
XGBoost R2=0.9650	LightGBM R2=0.9254	Total Records	275

Automated Financial Insights

■ Average Profit Margin is 16.2% — above the 15% industry benchmark.

■ 'Healthcare' is the best-performing sector. 'Finance' is the lowest-performing sector.

■ Average ROA is 8.16%. 97% of companies have positive ROA.

■ 26.5% of companies carry a Debt Ratio above 0.50, indicating elevated leverage risk.

■ Highest net profit company: Company_241.

■ Revenue-Net Profit correlation = 0.648 (moderate positive relationship).

■ Best ML Model: XGBoost - R2=0.9650, MAE=$33.7M, RMSE=$51.5M.

AI-Generated Narrative Report (Gemini AI)

EXECUTIVE SUMMARY

This report analyses 275 companies across 5 sectors (2019-2023). Average revenue is $2521.8M with net profit of $375.9M (16.2% margin).

SECTOR ANALYSIS

Technology and Healthcare show the strongest profitability. Energy and Retail face higher cost pressures and thinner margins.

PROFITABILITY & RISK

Average ROA is 8.16%. 27% of companies carry Debt Ratio > 0.50.

ML MODEL RESULTS

XGBoost R2=0.9650, LightGBM R2=0.9254. Best: XGBoost (R2=0.9650, MAE=$33.7M).

RECOMMENDATIONS

- Prioritise investment in high-ROA sectors (Technology, Healthcare)

- Implement debt restructuring for Debt Ratio > 0.5 companies

- Expand model with macroeconomic indicators

CONCLUSION

The portfolio shows resilient financial characteristics. AI-driven analytics will enhance decision accuracy and reporting speed.

ML Models: XGBoost & LightGBM

XGBoost and LightGBM are gradient-boosting tree ensembles trained to predict Net Profit from 18 financial features with early stopping to prevent overfitting. Feature importance charts reveal which financial metrics drive predictions most. Both models were compared on R2, MAE, and RMSE.

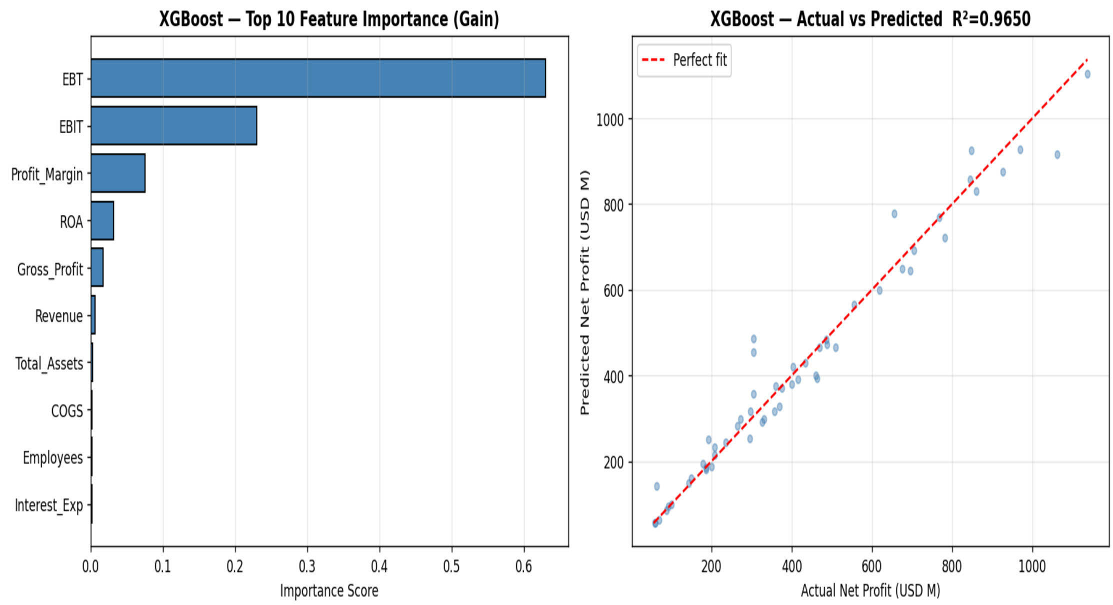

Figure A: XGBoost - Feature Importance & Actual vs Predicted

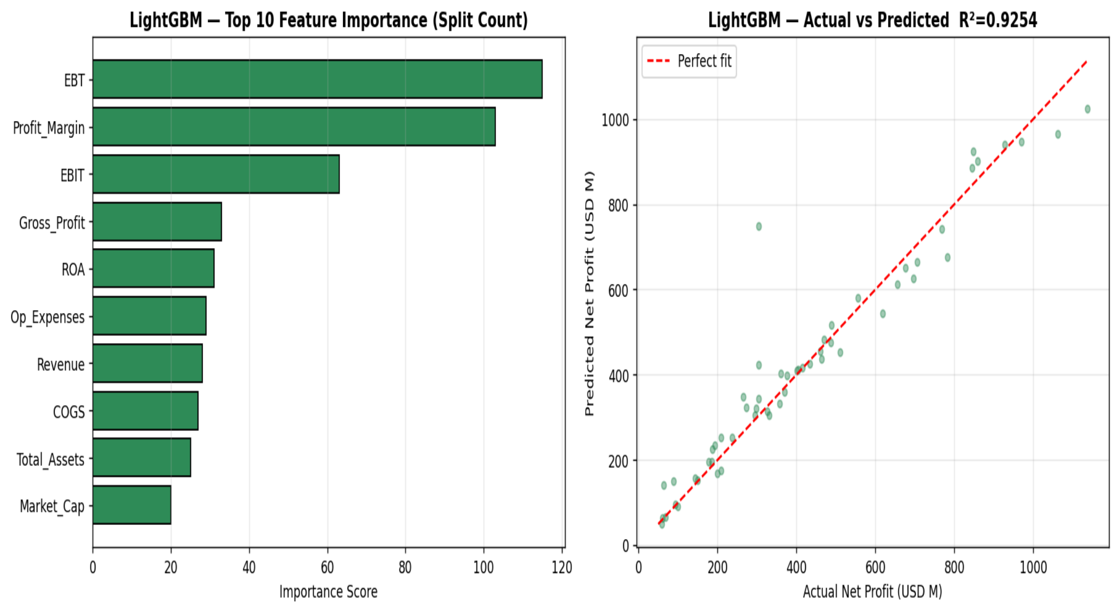

Figure B: LightGBM - Feature Importance & Actual vs Predicted

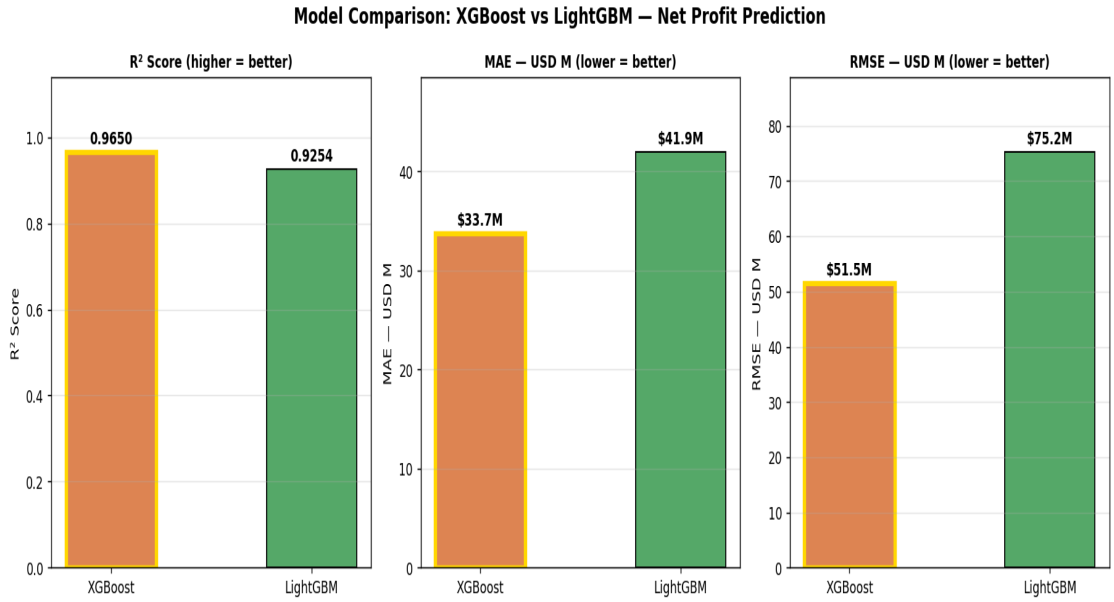

Figure C: XGBoost vs LightGBM - R2, MAE, RMSE Comparison

Data Visualizations

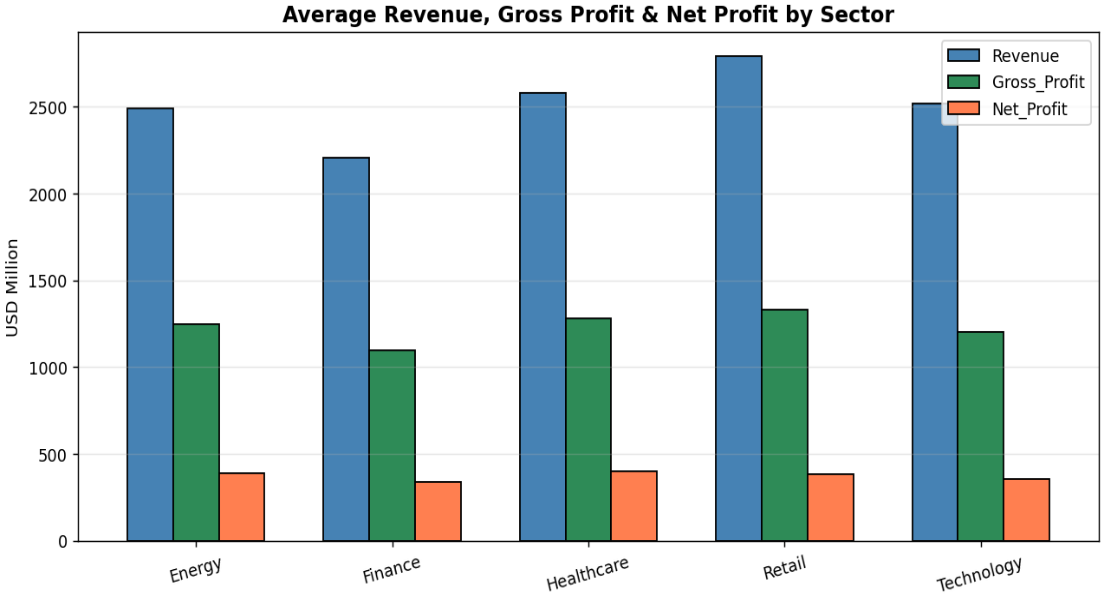

Figure 1: Avg Revenue, Gross Profit & Net Profit by Sector

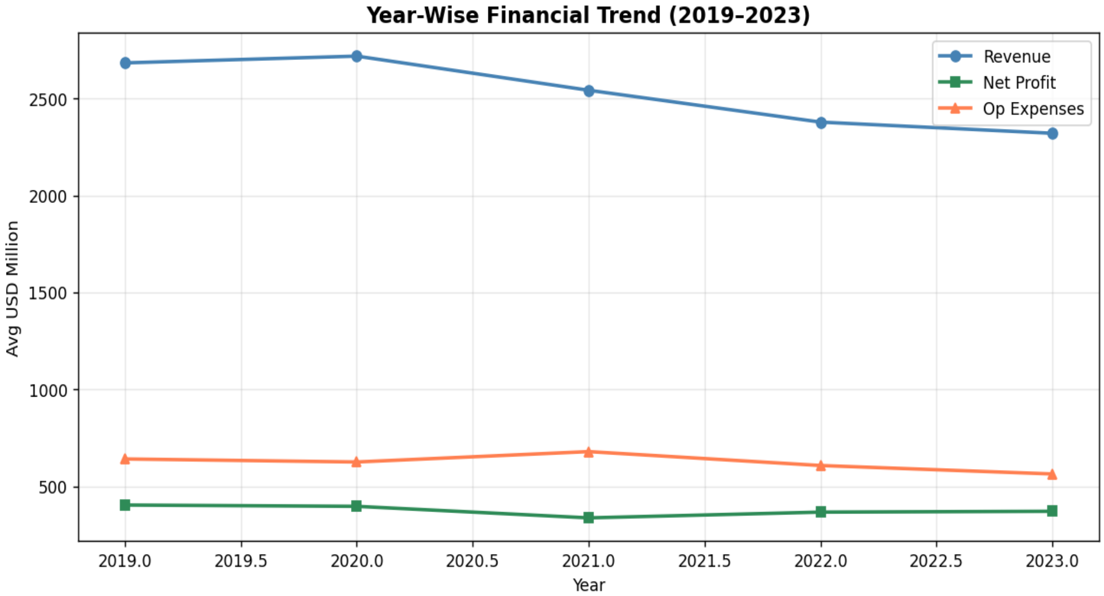

Figure 2: Year-Wise Financial Trend (2019-2023)

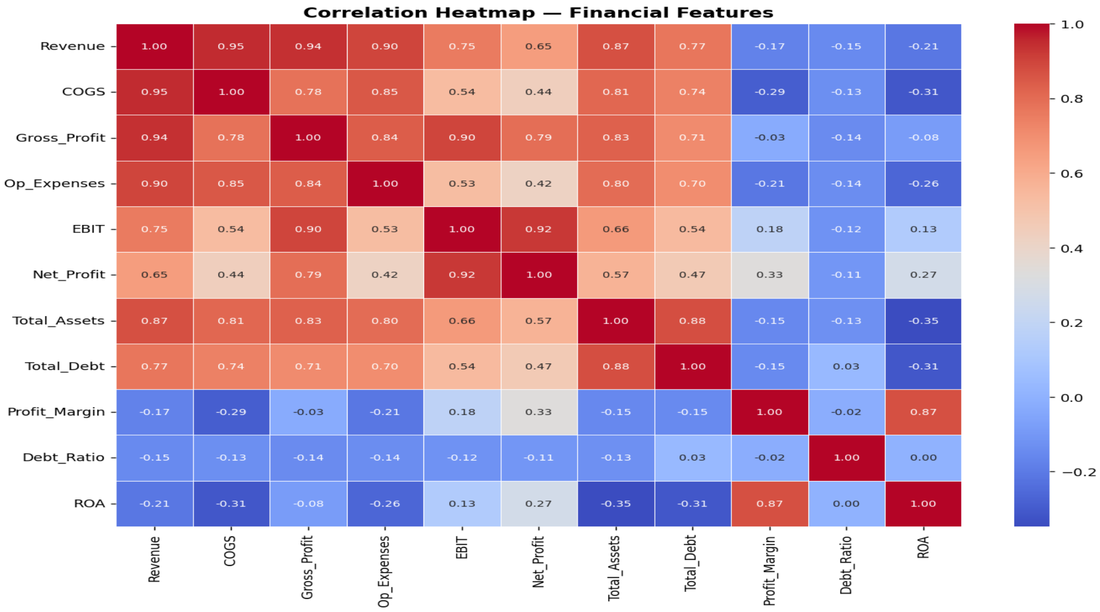

Figure 3: Correlation Heatmap - Financial Features

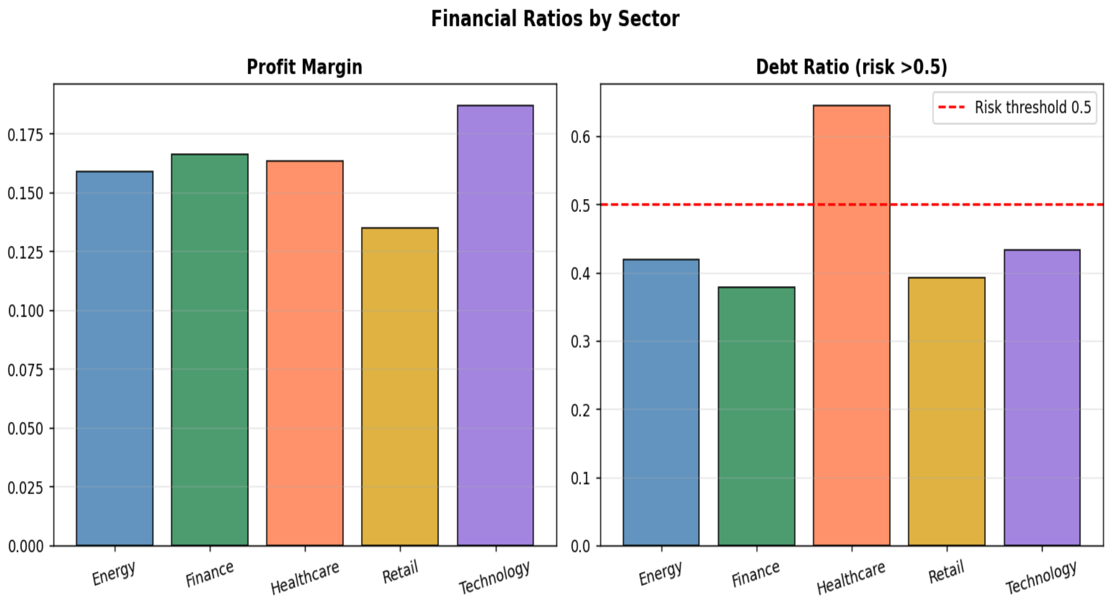

Figure 4: Profit Margin & Debt Ratio by Sector

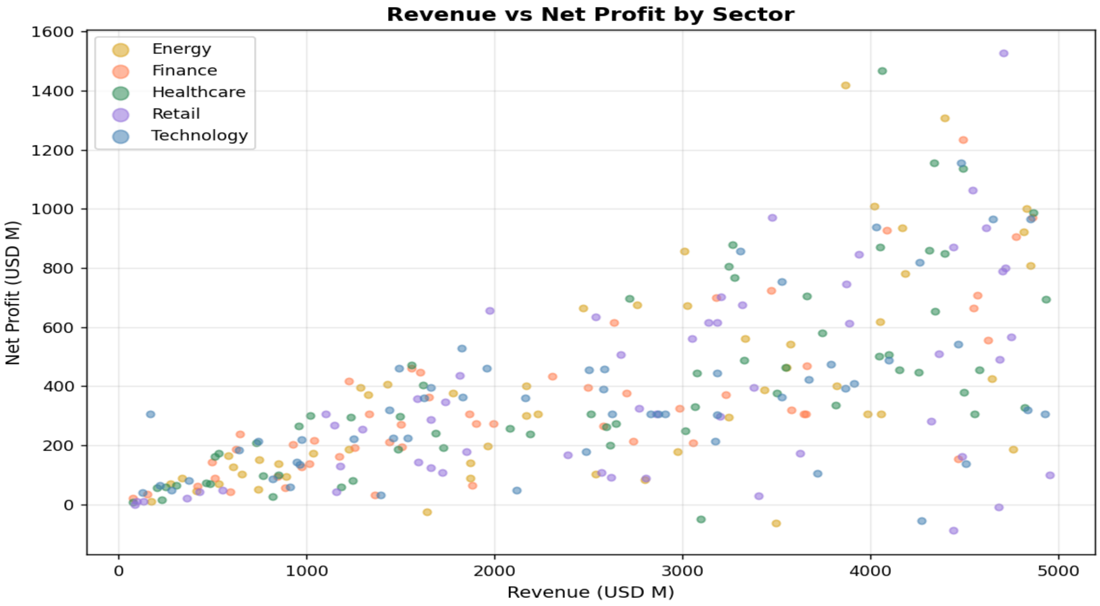

Figure 5: Revenue vs Net Profit by Sector

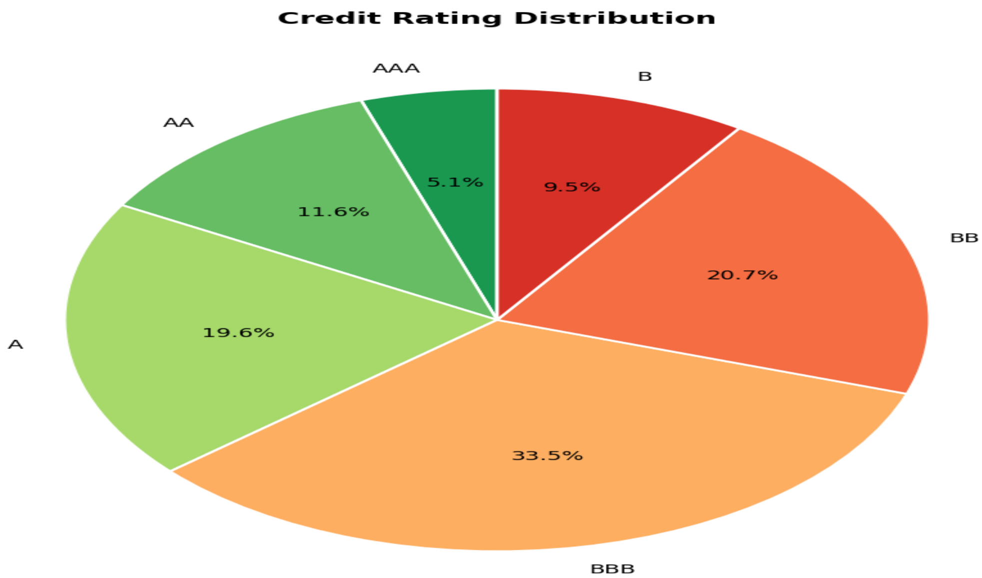

Figure 6: Credit Rating Distribution

Conclusion & Future Enhancements

■ End-to-end pipeline on 275 company records across 5 sectors and 5 years.

■ XGBoost R2=0.9650 vs LightGBM R2=0.9254. Best model: XGBoost (MAE=$33.7M).

■ Gemini AI automatically produced a professional narrative financial report.

■ Interactive HTML dashboard opens inline in Colab via data URI - no server required.

■ Future: Real-time market data integration, Streamlit deployment, anomaly detection.